From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

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IITGN ROBOTICS

PART A How does geometry govern compliance?

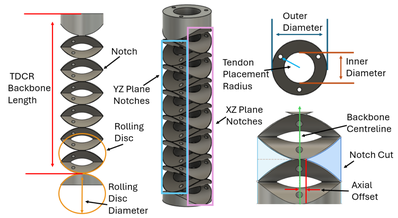
under review, ICRA 2027

The robot, and two questions

Printed notched tubes are cheap continuum robots for tight spaces, but there is no rule for choosing their notches and no live, sensor-free model of their shape.

A. Design Which geometry gives a wanted bend, and what stroke and force does it cost?

B. Model Can the whole shape be predicted live from servo commands alone, for any number of modules?



TPU tube with two notch families and tendons in the wall; 160 mm modules.

What is new

A SWEEP: CAD to FEM, unattended; 616 runs link geometry to bend and force.
B A twin from the CAD's sections, pull-only tendons and gravity: 1 to 4 ms.
B It learns the module's law, not the shape: fitted on 1–2 modules, it holds on 4.

A1 SWEEP: geometry to simulation, unattended

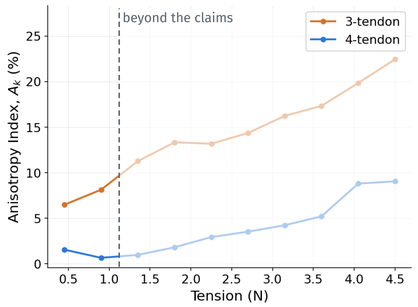
Parameters D_{RD}, L, D_o, D_i, e, c, N_t → **CAD** CadQuery, one Boolean cut → **Mesh** CGAL tetrahedra → **FEM** SOFA, pull-only tendons → **Metrics** curvature, CC error, stroke



designs	14
tendons × loads	4 × 11
FEM runs (done)	616 (614)
TPU modulus	56 MPa
gravity	on
claims valid for	$T \le 0.9$ N

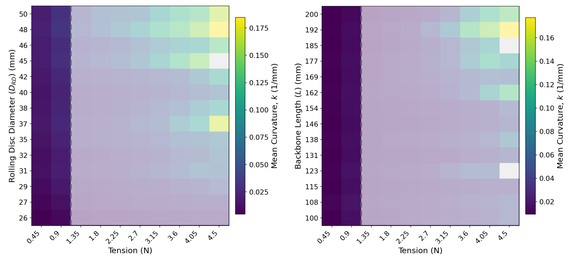
Same load, three disc diameters ($R_d = D_{RD}$): a larger disc packs more notches and bends further.

A2 Match the actuation to the structure



Three tendons bend unevenly with direction. Four cut the anisotropy $A_\kappa = \sigma_\kappa / |\bar{\kappa}|$ from 6.2 % to 2.6 % (13 of 14 designs) and need only two motors: two antagonistic pairs.

A3 What each parameter controls



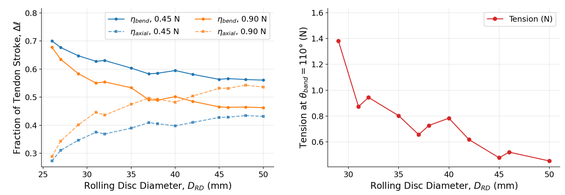
D_{RD} sets curvature turning 54° → 143° (26 to 50 mm, 0.9 N)

L sets reach tip travel 78 → 157 mm (100 to 200 mm)

Mean curvature (1/mm). More notches bend further but stray up to 5× further from constant curvature. Faded: above 0.9 N, beyond the claims; grey: no settled run.

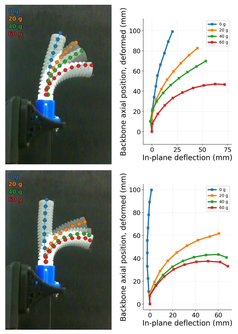
A4 Stroke budget: what a bend costs the actuator

$$\underbrace{\Delta\ell}_{\text{tendon stroke}} = \underbrace{\Theta\, d}_{\text{bending}} + \underbrace{\Delta L}_{\text{axial compression}}$$



The budget closes to 0.99; 42 to 47 % of the stroke only shortens the body (left). A larger disc halves the tension for a 110° bend, 0.87 to 0.45 N (right).

A5 Experiment: three printed prototypes



D_{RD} (mm)	NRMSE, % of length centreline	tip	axial
30	18.5	20.9	43.1
67.89	10.1	23.5	11.9
86.84	16.3	64.7	23.9

Shape agrees within 18.5 %; the tip less well (no friction or self-contact in the model). Prototypes span D_{RD} = 30 to 87 mm.

PART B A sensor-free, real-time digital twin

in preparation for IEEE RA-L

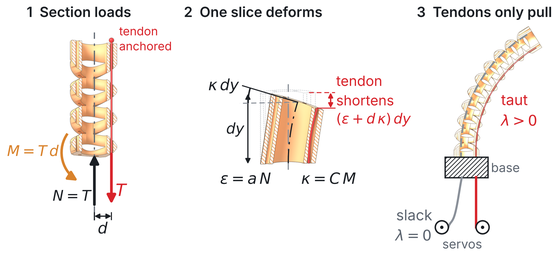
B1 Servo commands in, whole shape out

Servo commands $\delta\ell$ → **Tendon map** $\delta\ell = G\lambda$ from the CAD → **Pull-only QP** $\lambda \ge 0$ exact slack → **Gravity** fixed point CAD weight → **Shape** every 5 mm in 1 to 4 ms

its weight moves the tendon paths

No sensor on the robot. Module: the SWEEP design with D_{RD} = 30 mm, L = 150 mm. Laptop CPU, slowest call 26 ms; a Cosserat rod takes 360 ms.

B2 The physics: a beam that follows the CAD

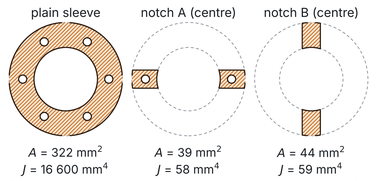


$$\delta\ell = G\lambda, \quad G_{ij} = \sum_c \tau_{ic}\tau_{jc}\left(a_c + \boldsymbol{d}_i^\top C_c\, \boldsymbol{d}_j\right) + c_t L_i\, \delta_{ij}$$

$$\lambda^\star = \arg\min_{\lambda \ge 0} \tfrac{1}{2}\lambda^\top G\lambda - \delta\ell^\top\lambda$$

Tendon map: exactly linear, any bend

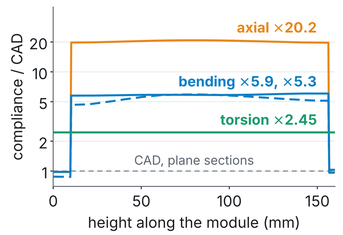
Pull-only: convex, unique, exact



A notch leaves two 4 mm bridges: 8 times less area, 280 times less bending stiffness. The notches do the bending.

$\delta\ell$: servo displacements; λ: tendon tensions. $\tau_{ic} \in \{0, 1\}$: tendon i passes slice c. $a_c = \int_c dy/A$, $C_c = \int_c dy/J$: slice compliances from the CAD. $c_t L_i$: tendon stretch.

B3 Identify the module law once, then verify



Fitted through the solver on 675 FEM poses of one and two modules: four scales on the CAD's compliance, plus a smooth correction within 13 %.

solver check	difference
closed-form beams	< 1 µm
Cosserat rod	0.004 mm
automated tests	132 pass

B4 Scientific ML: learn the law, not the solution

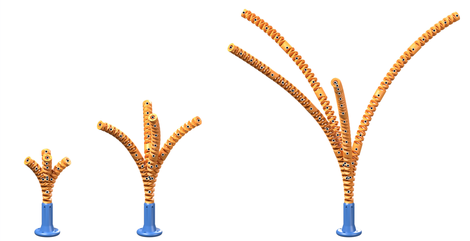
Learned law θ $C_\theta(y) = C_{CAD}(y)\, e^\theta$ 4 scales + smooth profile → **Exact physics** pull-only QP + gravity → **Shape** ⇄ loss → **Data** FEM now, mocap next

fit θ through the solver

tip error vs FEM (mm)	1 and 2 modules	4 modules (unseen)
learn the **solution** (PINNs)	3 to 6	54 to 73, retrained
learn the law, in the solver	**0.2 to 0.4**	**0.50, same θ**

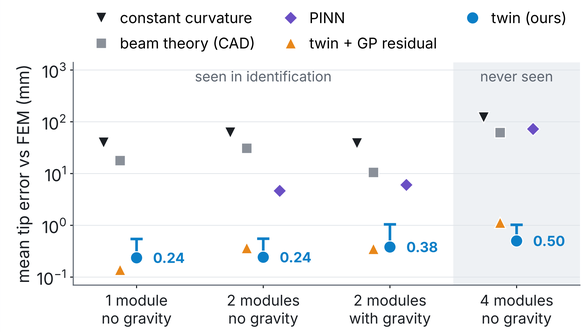
One θ for every number of modules. PINNs: forward and inverse, same data.

B5 Calibrate small, deploy large (FEM ground truth)



Fitted on one and two modules; on four, the error is **0.08 %** of length.

The twin rendered on the CAD, one scale. Dots: FEM centreline.



Mean and 95th percentile over held-out poses, bends to 75°. Constant curvature over-bends 1.7×: it ignores A4's compression. FEM meshes differ by 0.6 mm.

B6 Towards hardware: what the data can identify

problem (synthetic data)	remedy in the calibration
weight and stiffness confounded	weigh modules, fix the mass
equal pulls slacken distal tendons	pretension 1 : 1.75 : 2.25 : 2.5
0.4 mm moment-arm error	fit as 5th number, to 0.02 mm



Right: the parts designed for the four-module robot.

Conclusions

A. Design. The disc diameter sets curvature (54° → 143° at 0.9 N), the length sets reach, and 42 to 47 % of the stroke is compression.

B. Model. Identified on one and two modules, the twin predicts four to 0.50 mm (0.08 % of length) in 1 to 4 ms; a learned shape does not.

Limits. FEM only, at its resolution (about 0.6 mm); quasi-static; no friction or contact in the identification; bends up to 75°.

Builds on: Camarillo et al., T-RO 2008 • Rucker & Webster, T-RO 2011 • Faure et al., SOFA, 2012 • Coevoet et al., SoftRobots, 2017 • Kim et al., arXiv:2507.10121 • Danesh et al., arXiv:2608.26273.

Direction A geometry-conditioned twin for inverse design

1 Hardware validation (now)
Motion capture, 4-module robot. Calibrate **5 numbers on 15 poses** of 2 modules; test on 4 modules.
Pass: tip error ≤ 2 % of length (13 mm), fixed before any data.

2 Law as a function of geometry
SWEEP generates designs; the twin identifies each: $\theta = f(D_{RD}, D_i, L)$.
Pass: beats CAD-only theory on designs held out entirely.

3 Inverse design
Target curvature + stroke budget → geometry, in milliseconds; print and verify.
If residuals demand: convex notch energy ψ_θ, friction, self-contact.

Outputs and my role

[1] D. Yadav, **O. M. Pawar**, Krishnapriya R, S. Gandhi, M. Vadali, "How Geometry Governs Compliance? A Parametric Study of Rolling-Disc Notched Tendon-Driven Continuum Robot," **under review, IEEE ICRA 2027**.
[2] O. M. Pawar et al., sensor-free digital twin of a modular notched TDCR, in preparation for IEEE RA-L, with an installable open package.

Role. Part A: SOFA 25.06 port, parallel sweeps, mesh and settling checks. Part B: model, solver, identification, calibration, baselines and software.

Thanks to Divij Yadav, Krishnapriya R and Shaili Gandhi for SWEEP and the prototypes.

Report (PDF) Code (GitHub)